

ABHIJIT KUMAR GHOSH

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Portfolio – LinkedIn – Google Scholar – ResearchGate – ORCID

RESEARCH INTERESTS

Computer Vision – Medical Health AI – Health informatics – Deep Learning

EDUCATION

BRAC University

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering; CGPA: 3.63/4.00

May 2018 – January 2023

Department: Department of Computer Science and Engineering, School of Data and Sciences

Thesis: ML-Based Career Suggestive System for the Informal Job Sector Considering Cognitive Skills

Supervisor: Dr. Md. Khalilur Rahman, Professor, BRAC University

- Developed an ML-based career recommendation system using cognitive-skill data from the informal employment sector; evaluated six classification models, with Random Forest achieving the highest accuracy of approximately 62%.

ACADEMIC EXPERIENCE

European University of Bangladesh

Dhaka, Bangladesh

Lecturer, Department of Computer Science and Engineering

March 2025 – Present

- Courses Taught:** Computer Fundamentals, Object-Oriented Programming, Web Programming, Digital Logic Design, Software Engineering, Theory of Computing, Compiler Design, Artificial Intelligence, and Image Processing.

ELITE Research Lab LLC

United States (Remote)

Research Assistant

September 2026 – Present

- Supervise two research groups working on computer vision and medical image analysis, providing guidance on research methodology, model development, experimentation, and technical implementation.

PROFESSIONAL EXPERIENCE

SYMPHONY Softtech Ltd.

Software Developer

February 2024 – March 2025

- Developed and maintained ADEX and SHAMPAN-VAT applications using ASP.NET Core 6, MySQL, Razor Pages, and jQuery.

Square Informatix Limited

Programmer

August 2023 – February 2024

- Developed and maintained EQMS features using ASP.NET Core 6, Oracle Database, and AngularJS, including debugging and performance improvement.

JOURNAL MANUSCRIPTS

Submitted / Under-Review Journal Articles

- Abhijit Kumar Ghosh**, Ar Rafiul Islam Rafim, Fahmid Al Farid, Tanvir Ahmed, Enam Ahmed Taufik, Jia Uddin, and Hezerul Abdul Karim. “Gated Dual-Scale Feature Fusion for Colorectal Cancer Grading in Centralized and Federated Pathology.” Submitted to *Informatics in Medicine Unlocked*, 2026.
- Jannat-E-Nur, Enam Ahmed Taufik, **Abhijit Kumar Ghosh**, and Jia Uddin. “From Benchmark Success to Clinical Readiness in AI-Based Brain Tumour MRI Analysis: A Systematic Scoping Review.” Submitted to *Journal of Neuroradiology*, 2026.

CONFERENCE PUBLICATIONS AND MANUSCRIPTS

Published Conference Papers

- Md Shahriar Sajid, **Abhijit Kumar Ghosh**, and Fariha Nusrat. “PaperNet: Efficient Temporal Convolutions and Channel Residual Attention for EEG Epilepsy Detection.” In *Proceedings of the International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, Advances in Intelligent Systems Research, vol. 206, pp. 1159–1174, Atlantis Press, 2026. **DOI:** 10.2991/978-94-6239-664-7_79.
- Anika Islam, Md Ahasanul Arafath, **Abhijit Kumar Ghosh**, Tawsif Mustasin Rafin, Turjoy Das, Simantha Saha, Fateha Jannat Aydin, and Mizanur Rahman. “Tea Leaf Disease Classification and Out-of-Distribution Detection Using Deep Learning.” In *2025 IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE 2025)*, pp. 1032–1036, IEEE, 2025. **DOI:** 10.1109/WIECON-ECE69386.2025.11525958.
- Fateha Jannat Aydin, Mizanur Rahman, Ashifur Rahman, Vishwanath Akuthota, **Abhijit Kumar Ghosh**, Md Maruf Hasan Khondaker, Muntasim Ul Haque, Montaser Abdul Quader, and Md Tanzim Reza. “Multi-Agent Phishing Detection and Deletion via Small VLM and LLM Reasoning.” In *2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2026)*, IEEE, 2026. **DOI:** 10.1109/ICECTE69292.2026.11429303.

Conference Presentations

- Md Ahasanul Arafath, Md Tanzim Reza, Md Hasan Moon, Simantha Saha, Sadia Shobnom, Md Rony Ahmed, and **Abhijit Kumar Ghosh**. “Comparative Multivariate Time-Series Forecasting of IMU-Based Gait Data for Parkinson’s Disease Using VAR, LSTM, and GRU Models.” Paper presented at the *International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, Dhaka, Bangladesh, 12–13 December 2025. Paper ID: 10382.
- Fateha Jannat Aydin, Md Mizanur Rahman, Mozdaher Abdul Quader, Vishwanath Akuthota, Mehedi Hasan Jony, Md Maruf Hasan Khondaker, Nur A. Shawal, Montaser Abdul Quader, and **Abhijit Kumar Ghosh**. “Consensus with LLMs: CoT-Arbitrated Byzantine Fault Tolerance in Network Intrusion Detection.” Paper presented at the *International Conference on Intelligent Data Analysis and Applications (IDAA 2025)*, Dhaka, Bangladesh, 12–13 December 2025. Paper ID: 10791.
- Abhijit Kumar Ghosh**, Chowdhury Sajadul Islam, Mohammad Azim, Fariha Nusrat, Md Ahasanul Arafath, and Md Tanzim Reza. “Swin Transformer-Based Multi-Class Classification of Gallbladder Diseases: A Comparative Study with Lightweight CNNs.” Paper presented at the *2nd Undergraduate Conference on Intelligent Computing and Systems (UCICS 2026)*, Rajshahi, Bangladesh, 17–18 January 2026. Paper ID: 18.

Accepted Conference Papers

- Swagotam Malakar, Tanvir Limon, **Abhijit Kumar Ghosh**, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Beyond Confidence: A Leakage-Audited Selective-Prediction Study of Acid-Fast Bacilli Detection in Ziehl–Neelsen Microscopy.” Manuscript submitted to the *International Conference on Digital Image Computing: Techniques and Applications (DICTA 2026)*, 2026. Paper ID: 79.
- Md Faisal Sheikh, Tanvir Limon, Enam Ahmed Taufik, **Abhijit Kumar Ghosh**, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Towards Trustworthy Tuberculosis Detection: Subgroup Calibration of Object Detectors in Sputum-Smear Microscopy.” Manuscript submitted to the *International Conference on Digital Image Computing: Techniques and Applications (DICTA 2026)*, 2026. Paper ID: 80.
- **Abhijit Kumar Ghosh**, Ar Rafiul Islam Rafim, Rokeya Khatun, Moynul Islam Mahi, Montaser Abdul Quader, Sumaiya Tabassum Lamia, Anika Islam, Saud Yeahhea Al Rahul, and Sahal Bin Saad. “Adaptive Multi-View Transformer for Banknote Detection.” **Accepted** for presentation at the *2026 IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)*.

ACADEMIC SERVICE

- **Final-Year Project Supervisor, European University of Bangladesh** 2025–Present
 - **Automated X-ray Bone Fracture Classification Using Deep Learning Techniques**
Supervised students Joba Kundu, Md. Ibrahim Mia, Jabin Tasmin, Aminur Rahman, and Sulaiman Hossain in developing and evaluating CNN, transfer-learning, and vision-transformer models for X-ray-based bone-fracture classification.
 - **Smart Schedule Maker: A Hybrid Approach Using Heuristic Search with Constraint-Based Resource Allocation and Scheduling for Academic Institutions**
Supervised Md. Ahasanul Arafath in developing a Django and PostgreSQL-based academic scheduling system using constraint satisfaction, heuristic backtracking, and human-in-the-loop conflict resolution.
- **Member Secretary, AI Club, European University of Bangladesh** Spring 2025–Present
 - Coordinate academic and administrative activities, including AI workshops, seminars, coding competitions, and research-oriented events.
 - Mentor students in artificial intelligence, machine learning, project development, and undergraduate research while supporting industry-academia collaboration.
- **Reviewer, Deep Learning IndabaX Nigeria** 2026
 - Reviewed research submissions in Computer Vision and Multimodal AI, AI for Health and Biomedical Applications, and Distributed, Federated, and Privacy-Preserving AI.
 - Evaluated manuscripts based on originality, technical rigor, methodological quality, and practical relevance.
- **Reviewer, International Conference on Machine Intelligence and Network Data Security (MINDs 2026)** 2026
 - Review submissions in biomedical engineering, healthcare informatics, AI, machine learning, data analytics, computer vision, medical imaging, and related intelligent systems.
 - Evaluate manuscripts based on originality, technical rigor, methodological quality, relevance, and research contribution.
 - IEEE Conference Record ID: 69402; NSHM Knowledge Campus, Durgapur; 19–20 November 2026.

SELECTED PROJECTS

AI-Based Grade Prediction System

- Built a machine-learning student-performance prediction system using quizzes, attendance, assignments, and participation indicators.
- Used Python, scikit-learn, and SQL Server to analyze historical student data and identify at-risk learners.

Bangla Language Developing System

- Proposed a Bangla language learning and assessment platform for educational and job-sector examinations in Bangladesh.

Fire Alarm System in Farm

- Developed an Arduino Nano-based early-warning system for fire hazards in farm environments.

HONORS AND ACADEMIC RECOGNITION

- Vice Chancellor’s List Honoree for outstanding academic performance in three semesters.
- Dean’s List Honoree for academic excellence in one semester.

TECHNICAL SKILLS

Programming:	Python, Java, C++, C#
ML Frameworks:	PyTorch, TensorFlow, scikit-learn, OpenCV
Research Methods:	Data preprocessing, feature engineering, model development, experimental design, performance evaluation, statistical analysis
Databases:	MySQL, Oracle, SQL Server
Web / Software:	HTML, CSS, JavaScript, Apache Web Server
Tools:	LaTeX, Git, Jupyter Notebook, Google Colab
Languages:	Bangla (fluent), English (fluent), Chinese (professional proficiency)